



Massachusetts Institute of Technology

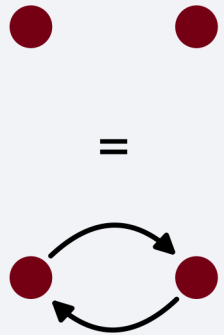
Anyonic Neural Network Wavefunctions

A.P. Fadon, P.M. Tam, **L. Shackleton**, arXiv:2609.xxxxx

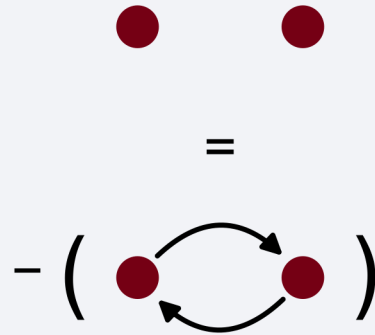
L. Shackleton, arXiv:2610.xxxxx

L. Shackleton, T. Wang, M. Metlitski, arXiv:26??.xxxxx

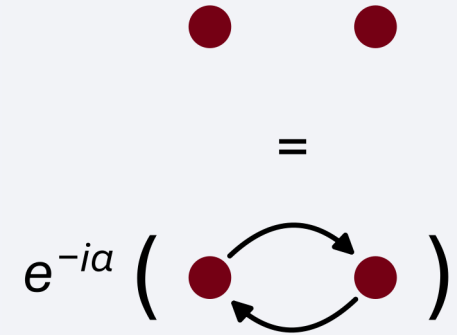
Generalized statistics in anyonic particles



Bosons

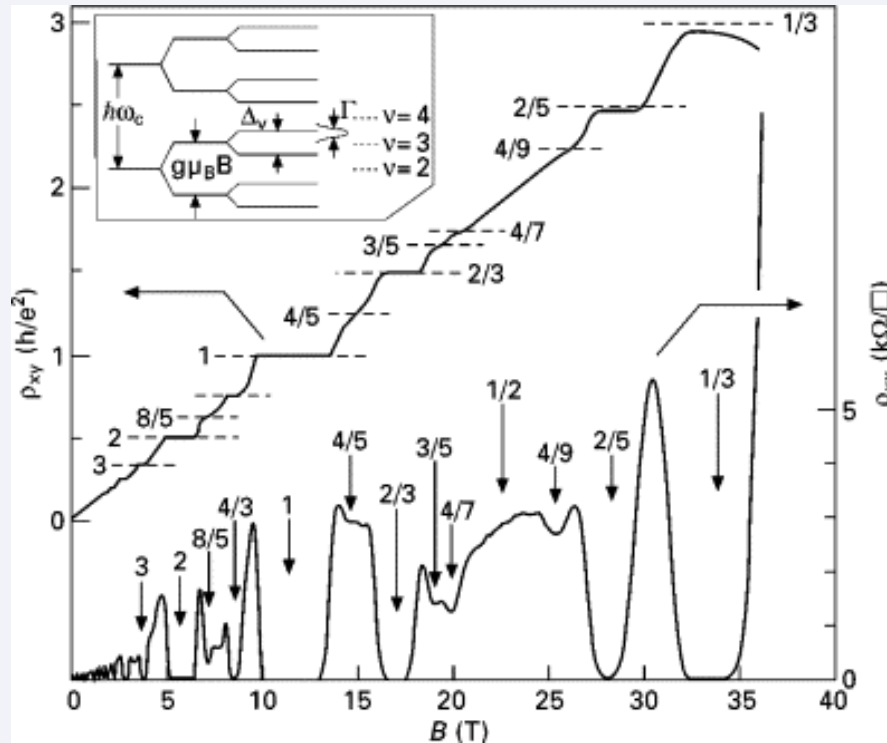


Fermions

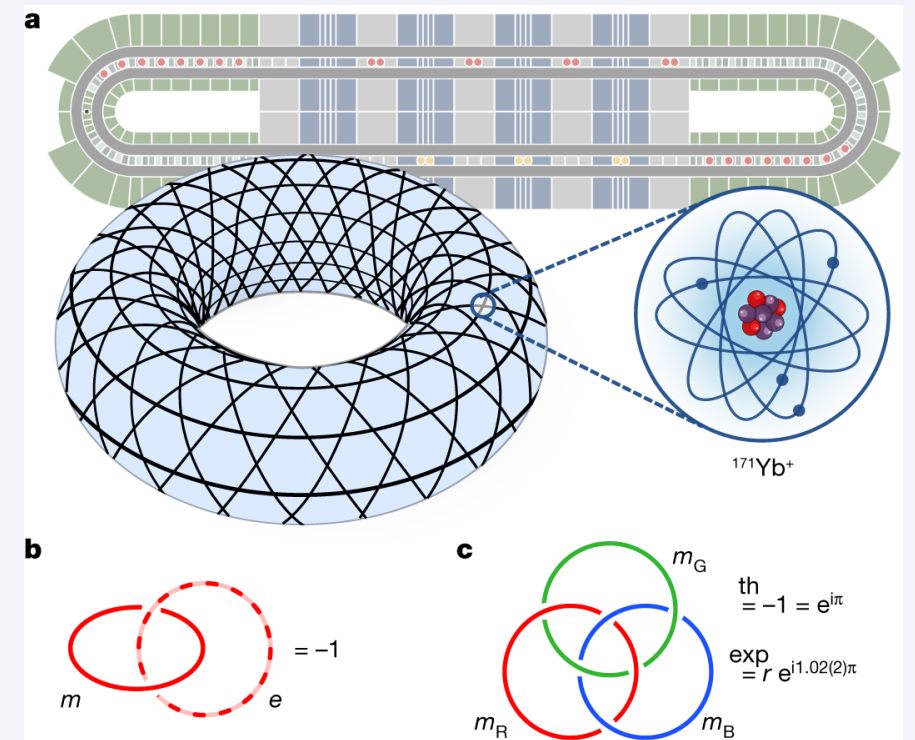


Anyons

Anyons can be realized in physical systems

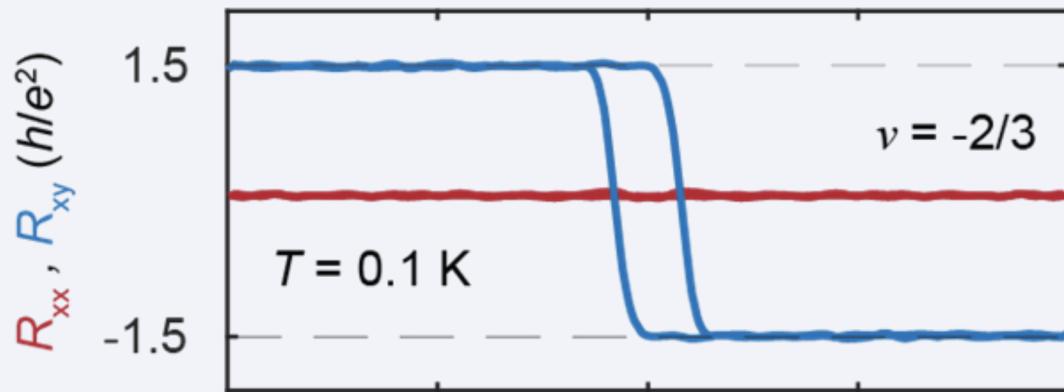


Fractional Quantum Hall Effect
Lai et al '04

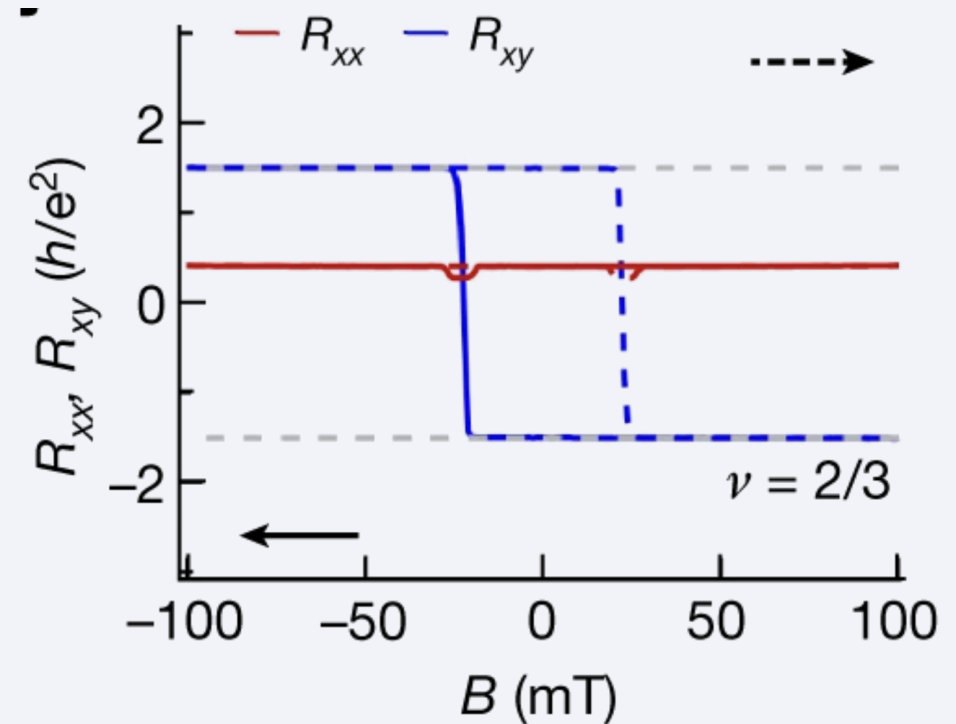


Simulations on quantum computers
Xu et al '24

Renewed interest from FQAH



Twisted MoTe2
Park et al, '23



Multilayer graphene
Lu et al, '24

No magnetic fields, anyons acquire **dispersion**

Numerous proposed theoretical outcomes

Shi and Senthil, '25

Zhang, '26

Nosov, Han, and Khalaf, '26

Shi and Senthil, '26

Pichler, Kuhlenkamp, Knap, and Vishwanath,
'26

Shi, Zhang, and Senthil, '25

Lotrič and Simon, '26

Wang, '26

Senthil, '26

Han, Vishwanath, and Khalaf, '26

Mehta, Nakajima, and Goldman, '26

...

Many superconductors

Holon metal

Vortex glass

Holon metal

Charge density wave

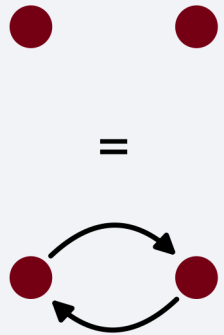
Pair density wave

Anyon crystal

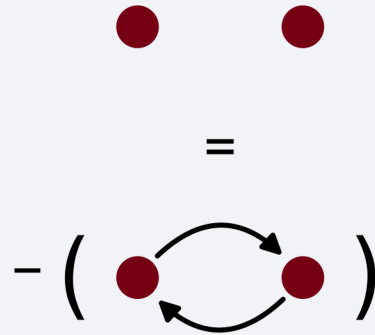
...

Outcome will likely depend on **microscopic details**

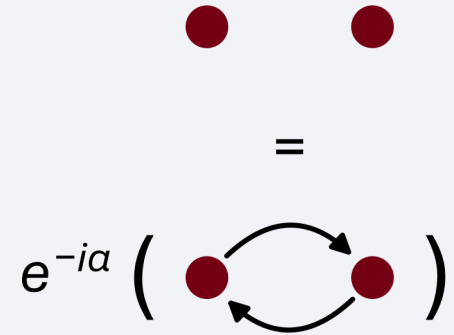
What are the effects of statistics at finite density?



Bose condensation



Pauli exclusion,
Fermi gas



??????

”Free” anyons are strongly interacting

$$H = -\frac{1}{2m} \sum_i \nabla_i^2 \quad \psi(\mathbf{r}_1, \dots, \mathbf{r}_i, \mathbf{r}_j, \dots, \mathbf{r}_N) = e^{i\alpha} \psi(\mathbf{r}_1, \dots, \mathbf{r}_j, \mathbf{r}_i, \dots, \mathbf{r}_N)$$

Solvable for two anyons (Arovas et al ‘85)

$$H = -\frac{1}{2m} \sum_i [\nabla_i - i\mathbf{a}_i]^2 \quad \psi(\mathbf{r}_1, \dots, \mathbf{r}_i, \mathbf{r}_j, \dots, \mathbf{r}_N) = \psi(\mathbf{r}_1, \dots, \mathbf{r}_j, \mathbf{r}_i, \dots, \mathbf{r}_N)$$

$$\mathbf{a}_i \equiv \alpha \sum_{j \neq i} \frac{\hat{\mathbf{z}} \times (\mathbf{r}_i - \mathbf{r}_j)}{|\mathbf{r}_i - \mathbf{r}_j|^2} \quad \nabla \times \mathbf{a}_i = 2\pi\alpha \hat{\mathbf{z}} \sum_{j \neq i} \delta^2(\mathbf{r}_i - \mathbf{r}_j)$$

An old idea: superfluidity

Smear out magnetic field, $\nabla \times \mathbf{a} \approx 2\pi\alpha\rho$

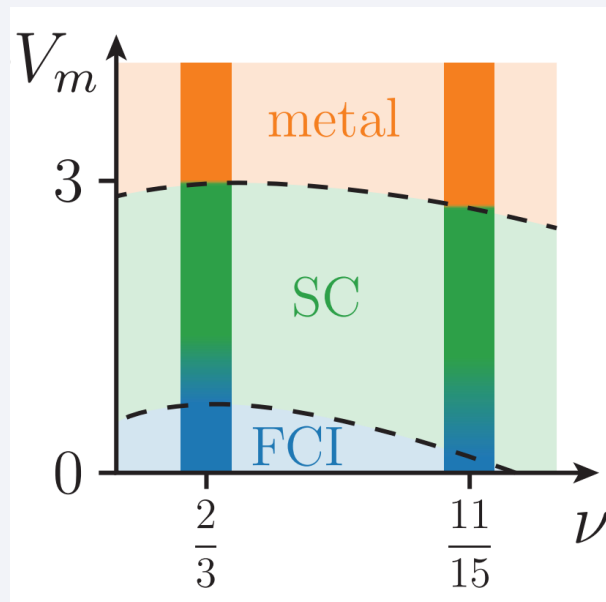
Near the fermionic limit, $\alpha = 1(1 - 1/n)$

Landau level filling: $\nu = \frac{2\pi\rho}{b} = n$

Flux responds to changing density: compressible

Anyons are difficult to simulate numerically

Anyons in FQH



Wang and Zaletel, '25

Anyon gas

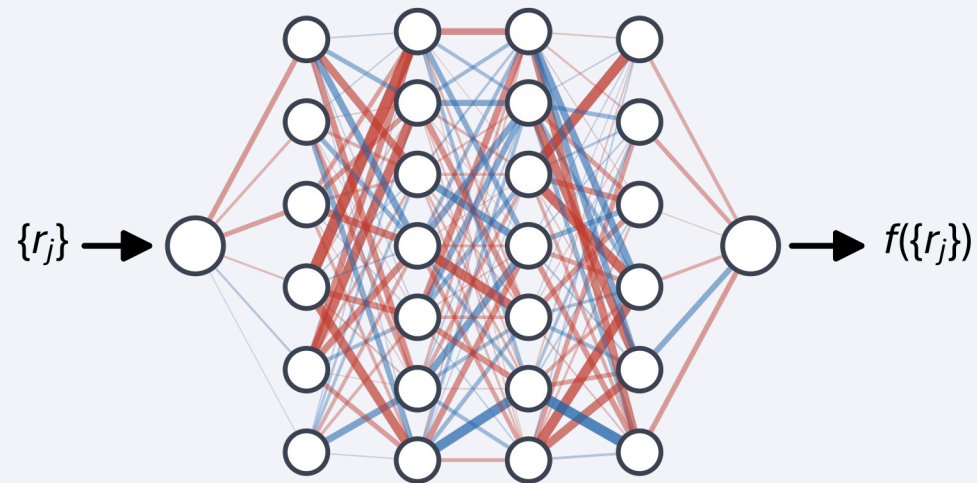
- Harmonic trap studied to $N = 4$ (Sporre et al, '92)
- Sphere, $N = 8$ (Xie et al, '90)
- Lattice models on a torus, $N = 6$ on 4×5 (Kallin, '93)

Neural quantum states are a significant advancement in our computational capabilities

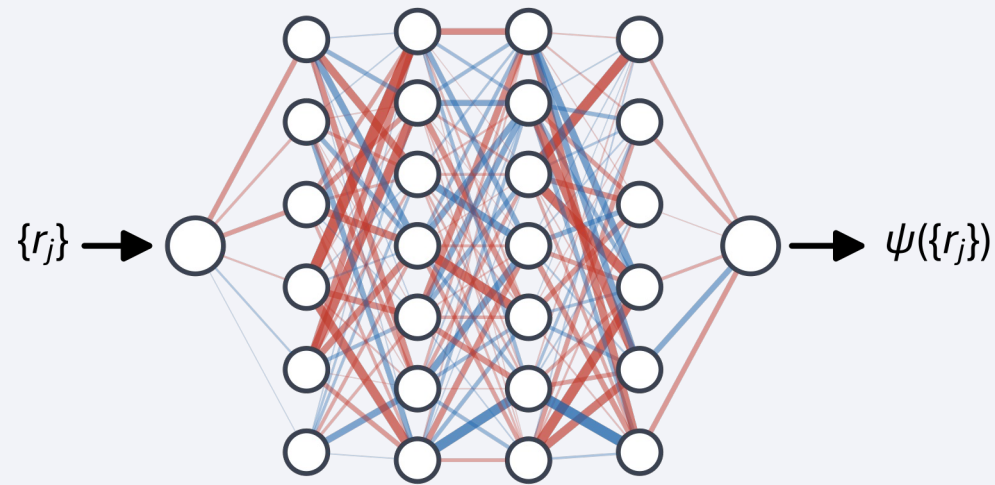
These methods are uniquely suited for tackling the anyon gas

Neural networks are powerful

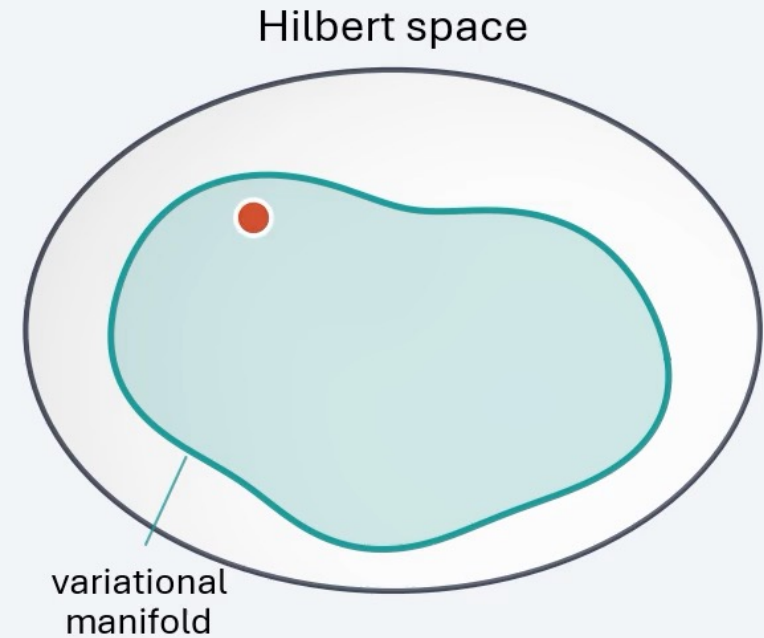
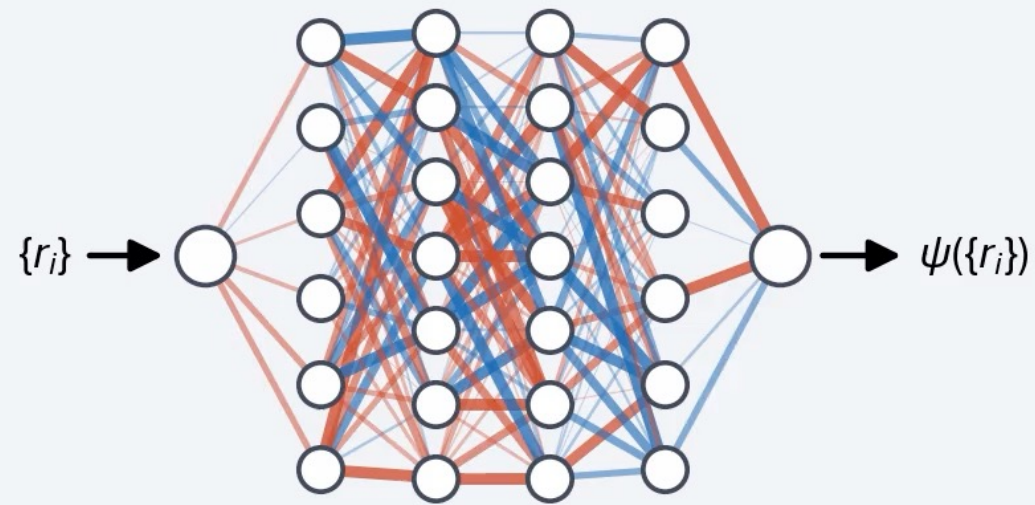
Neural networks are powerful **function approximators**



Neural networks are powerful **function approximators**



Neural Network Variational Monte Carlo



Neural Network Variational Monte Carlo

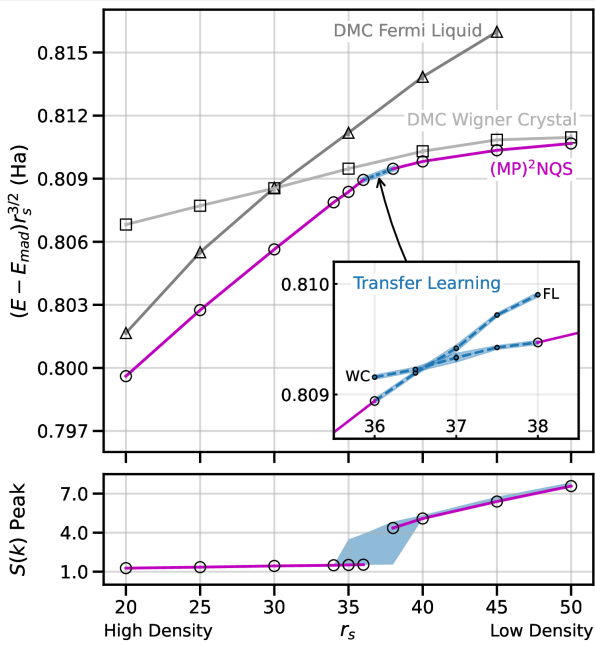
$$\frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{\int |\psi(x)|^2 \frac{H\psi(x)}{\psi(x)}}{\int |\psi(x)|^2} = \frac{\int p(x) \frac{H\psi(x)}{\psi(x)}}{\int p(x)}$$

No sign problem, can evaluate observables
as long as we can evaluate $\psi(x)$

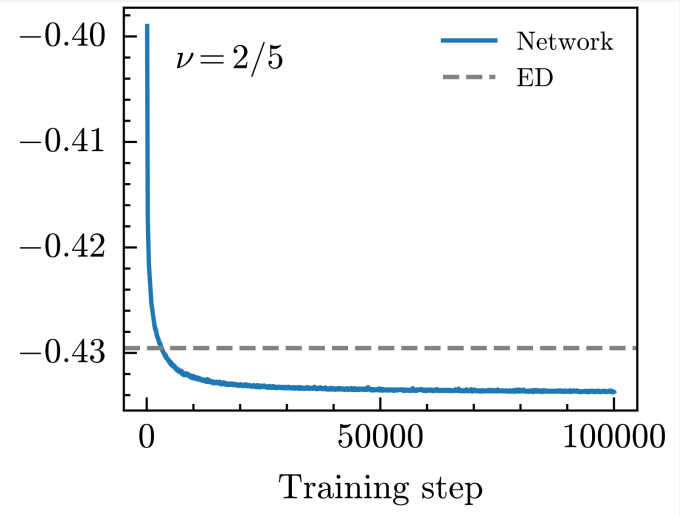
Procedure: evaluate $\frac{dE}{d\theta}$, update θ , repeat

NNVMC has stood out in several applications

Continuum systems



2D electron gas
Smith et al '24



FQH
Qian et al '25

Frustrated Magnetism

Table 1 | Ground-state energy on the 10×10 square lattice at $J_2/J_1 = 0.5$

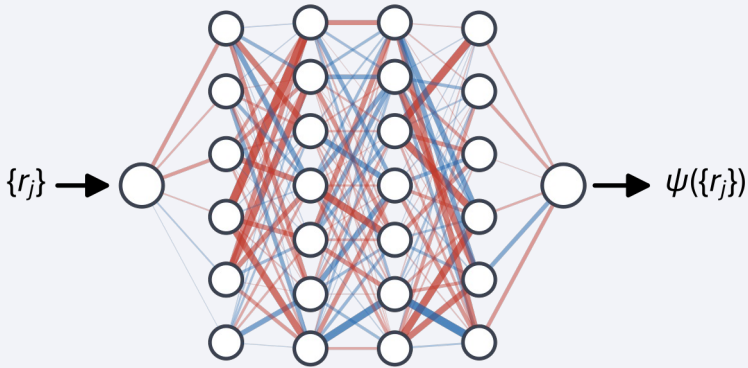
Energy per site	Wave function	# parameters
-0.48941(1)	MLP	893994
-0.494757(12)	CNN	Not available
-0.4947359(1)	Shallow CNN	11009
-0.49516(1)	Deep CNN	7676
-0.495502(1)	PEPS + Deep CNN	3531
-0.495530	DMRG	8192 SU(2) states
-0.495627(6)	aCNN	6538
-0.49575(3)	RBM-fermionic	2000
-0.49586(4)	CNN	10952
-0.4968(4)	RBM ($p = 1$)	Not available
-0.49717(1)	Deep CNN	106529
-0.497437(7)	GCNN	67548
-0.497468(1)	Deep CNN	421953
-0.4975490(2)	VMC ($p = 2$)	5
-0.497627(1)	Deep CNN	146320
-0.497629(1)	RBM + PP	13132
-0.497634(1)	Deep ViT	267720

Square lattices
Rende et al '24

Advantages in NNVMC

Versatile, many architectures

“Pure” neural network wavefunction



Carleo & Troyer, ‘17

Lower scale adaptations,
i.e. neural backflow

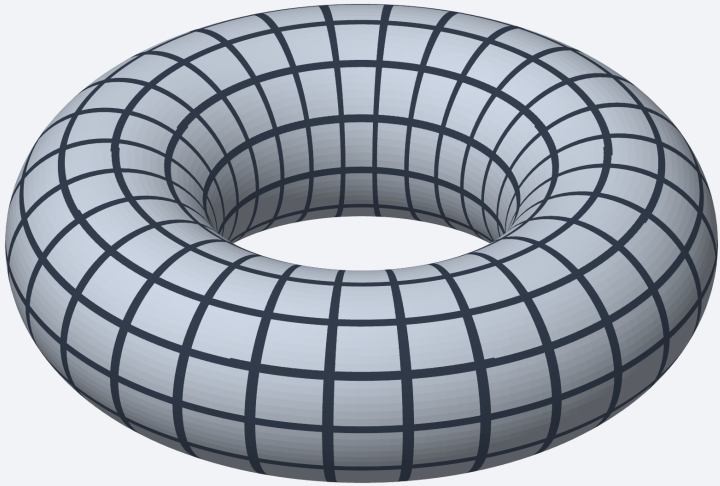
$$\psi(\{\mathbf{r}\}) = \det[\Phi]$$
$$\Phi_{ij} = \phi_i(\mathbf{r}_j) + f_{ij}(\{\mathbf{r}\})$$

Luo & Clark, ‘19

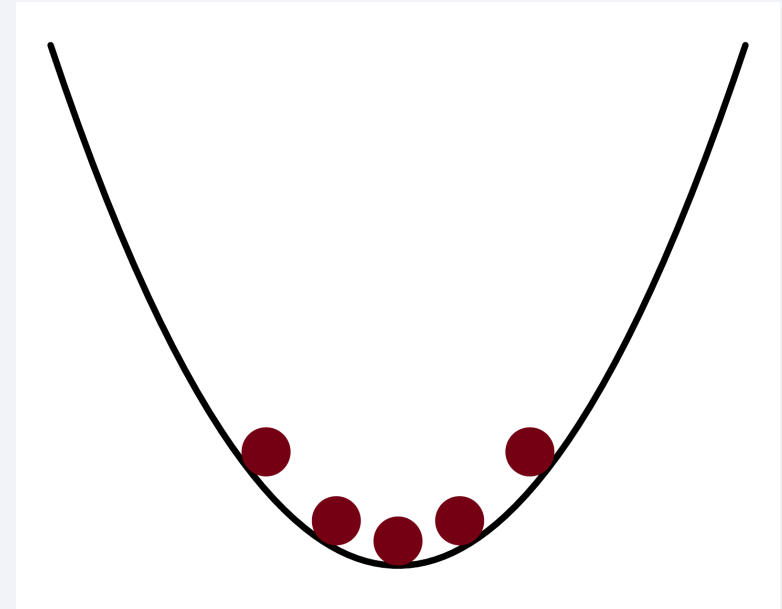
Efficient and scalable

i.e. 42x42 lattices for spin systems
Viteritti et al ‘26

Two systems of interest

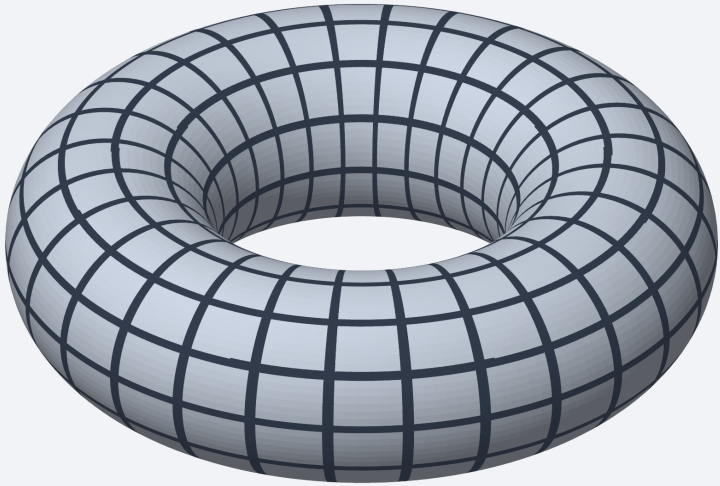


Lattice anyons on a torus

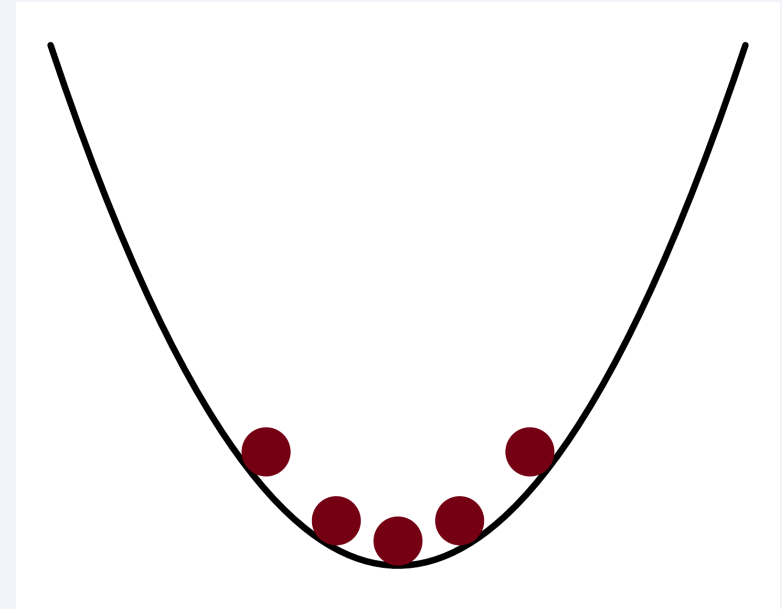


Continuum anyons in a harmonic trap

Two systems of interest

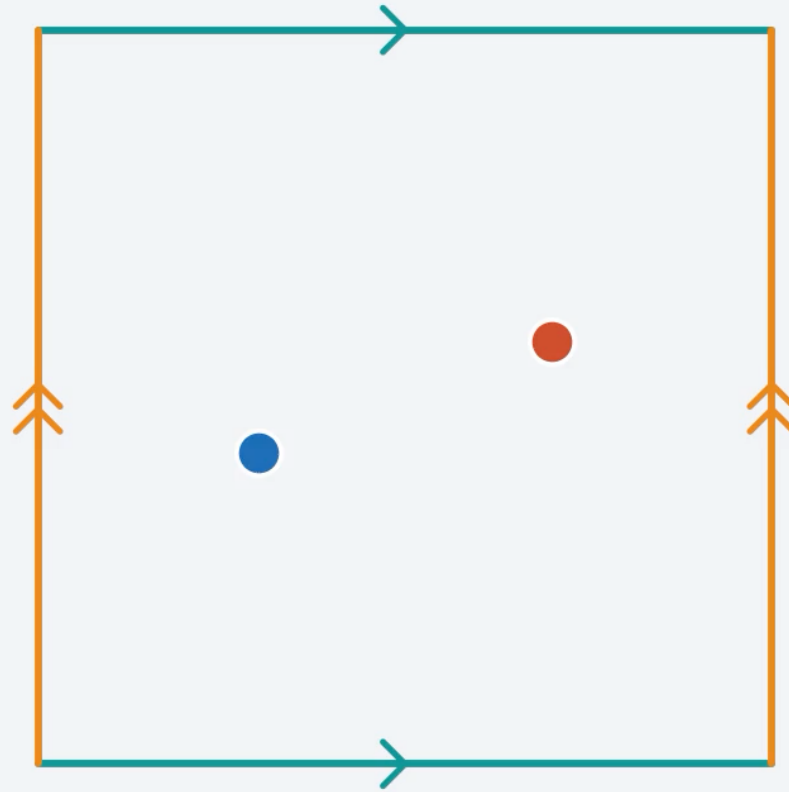


Lattice anyons on a torus



Continuum anyons in a harmonic trap

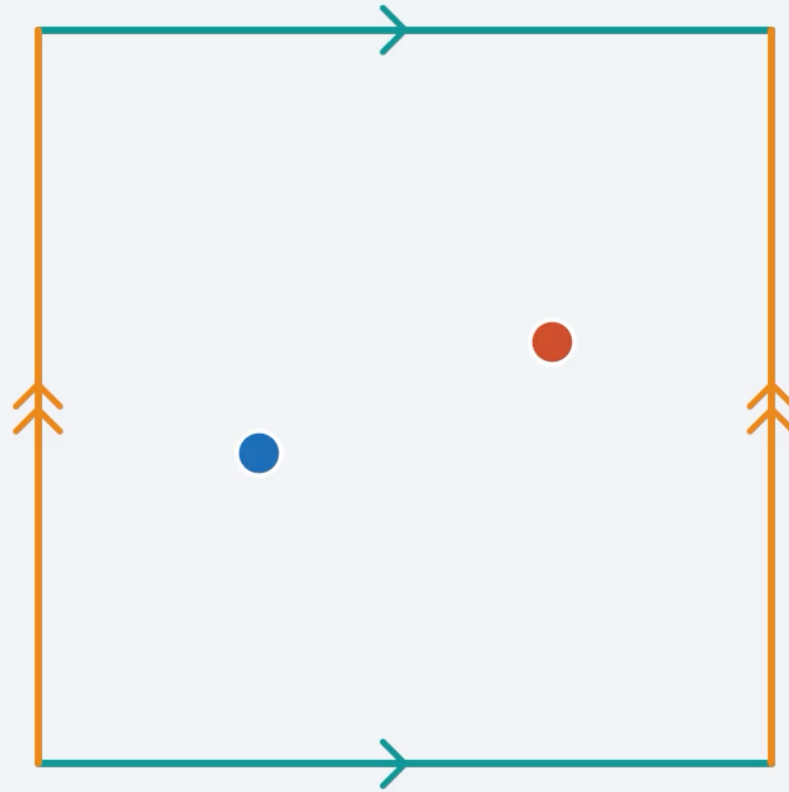
Obstacles to fractional statistics on a torus



T. Einarsson, Phys. Rev. Lett. **64**, 1995

X.G. Wen, E. Dagotto, E. Fradkin, Phys. Rev. B **42**, 6110

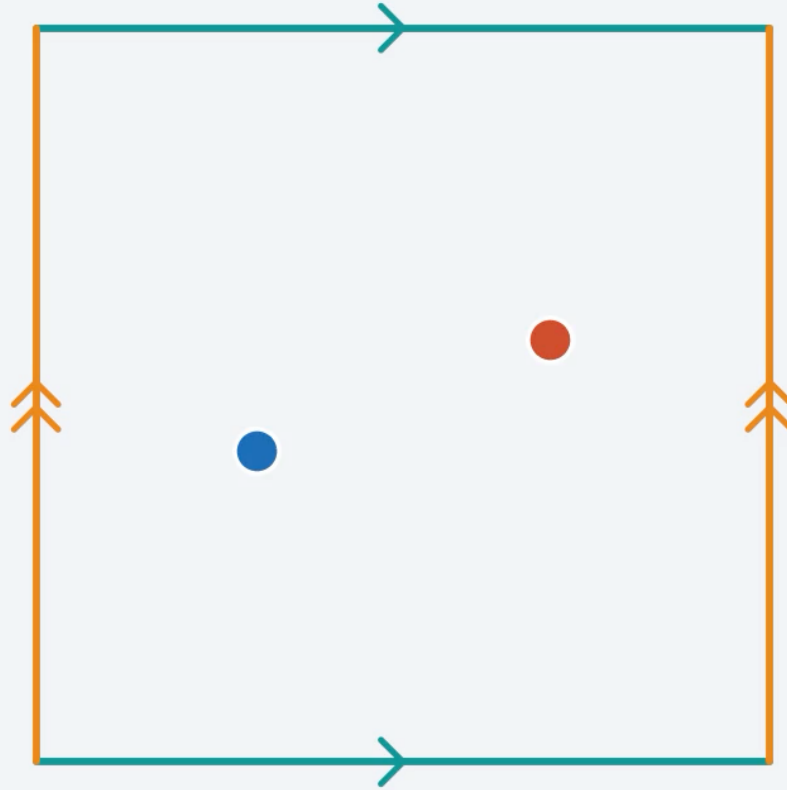
Obstacles to fractional statistics on a torus



T. Einarsson, Phys. Rev. Lett. **64**, 1995

X.G. Wen, E. Dagotto, E. Fradkin, Phys. Rev. B **42**, 6110

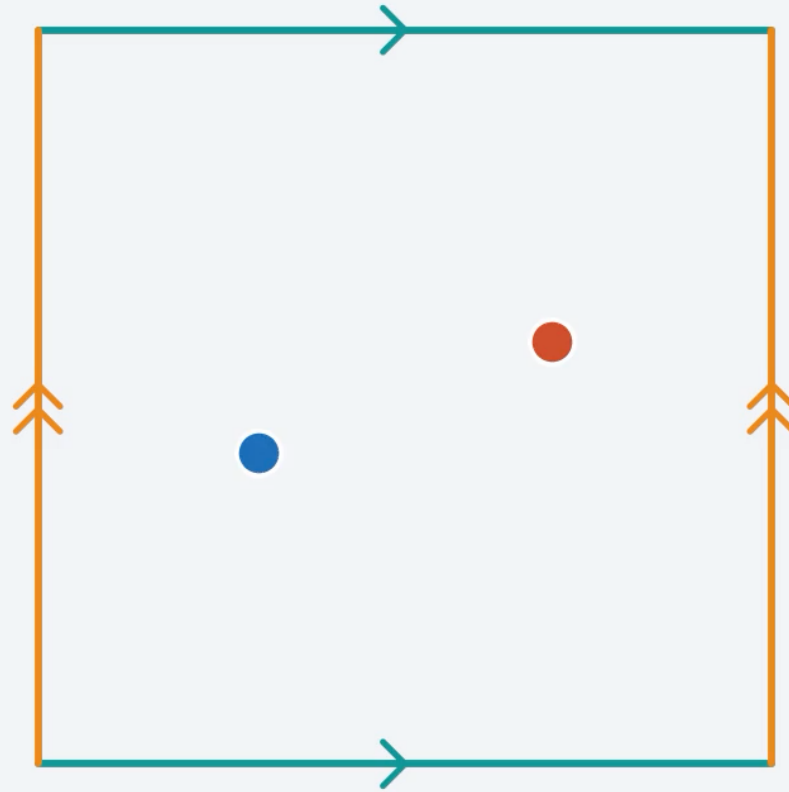
Obstacles to fractional statistics on a torus



T. Einarsson, Phys. Rev. Lett. **64**, 1995

X.G. Wen, E. Dagotto, E. Fradkin, Phys. Rev. B **42**, 6110

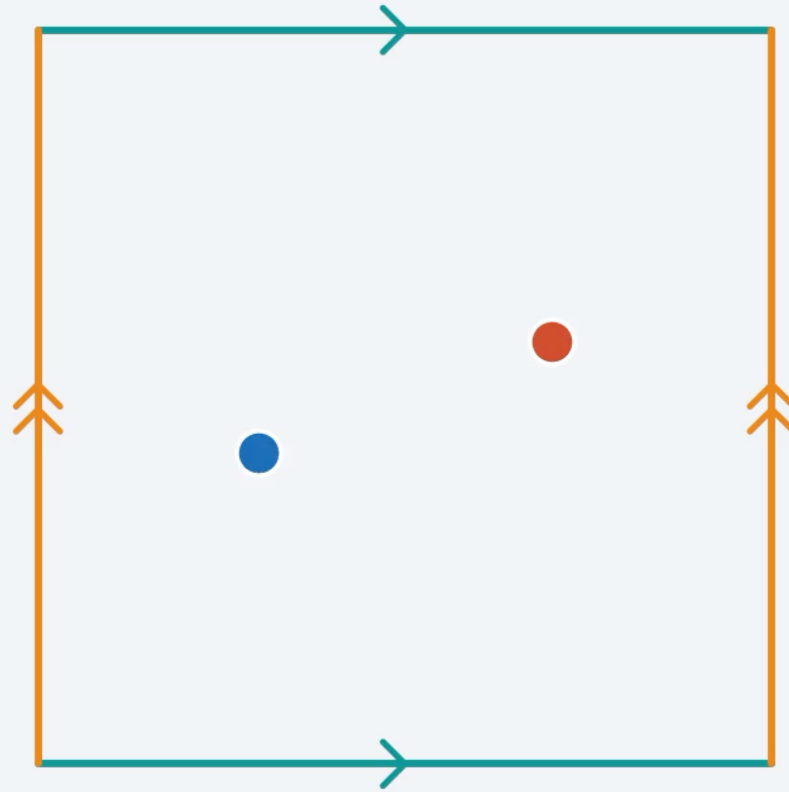
Obstacles to fractional statistics on a torus



T. Einarsson, Phys. Rev. Lett. **64**, 1995

X.G. Wen, E. Dagotto, E. Fradkin, Phys. Rev. B **42**, 6110

Obstacles to fractional statistics on a torus



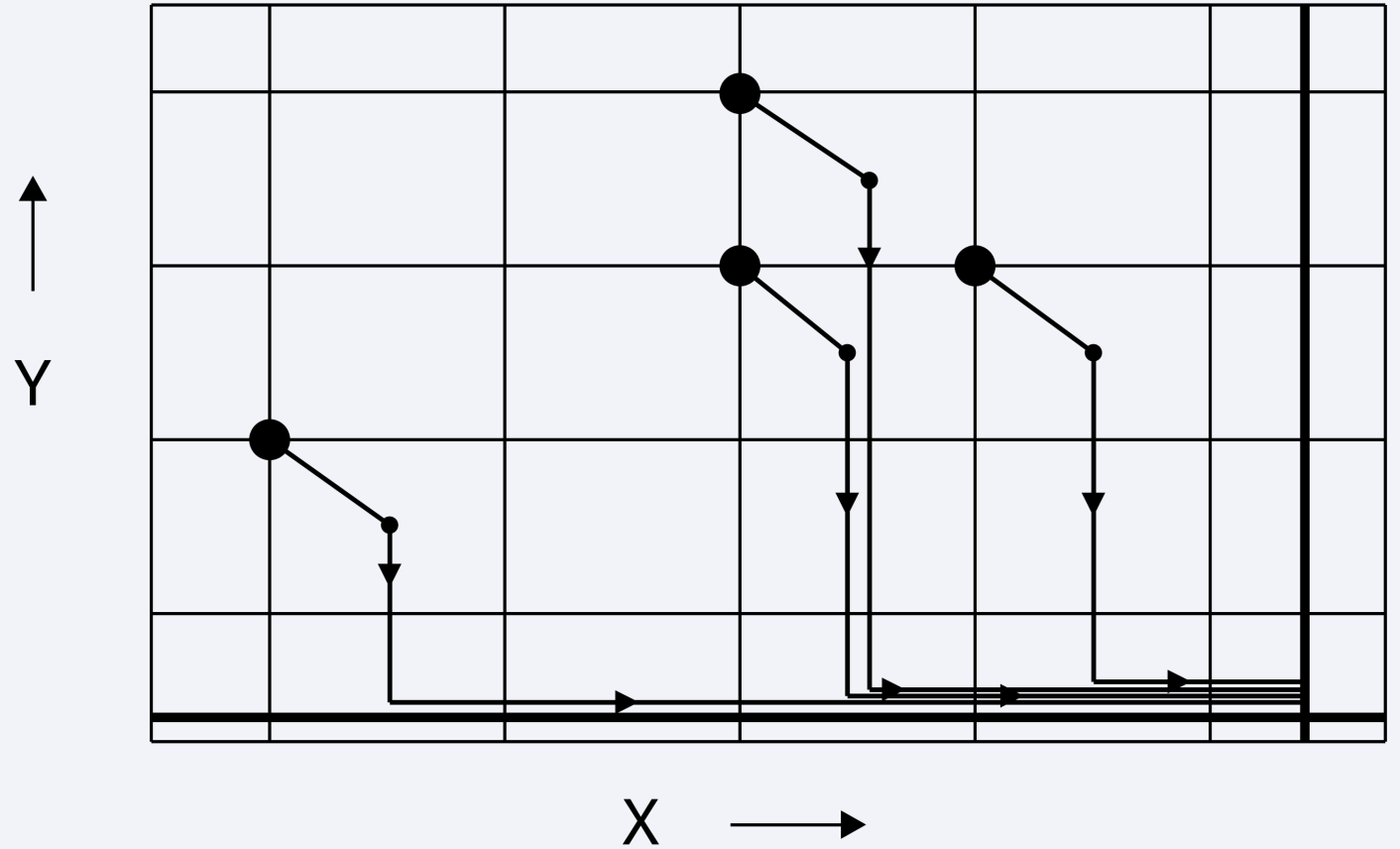
T. Einarsson, Phys. Rev. Lett. **64**, 1995

X.G. Wen, E. Dagotto, E. Fradkin, Phys. Rev. B **42**, 6110

Anyons require multi-component wavefunction

$$|\psi\rangle = |\ell_1, \ell_2, \dots, \ell_N\rangle \otimes |\sigma\rangle$$

Winding around torus
acts non-trivially on σ

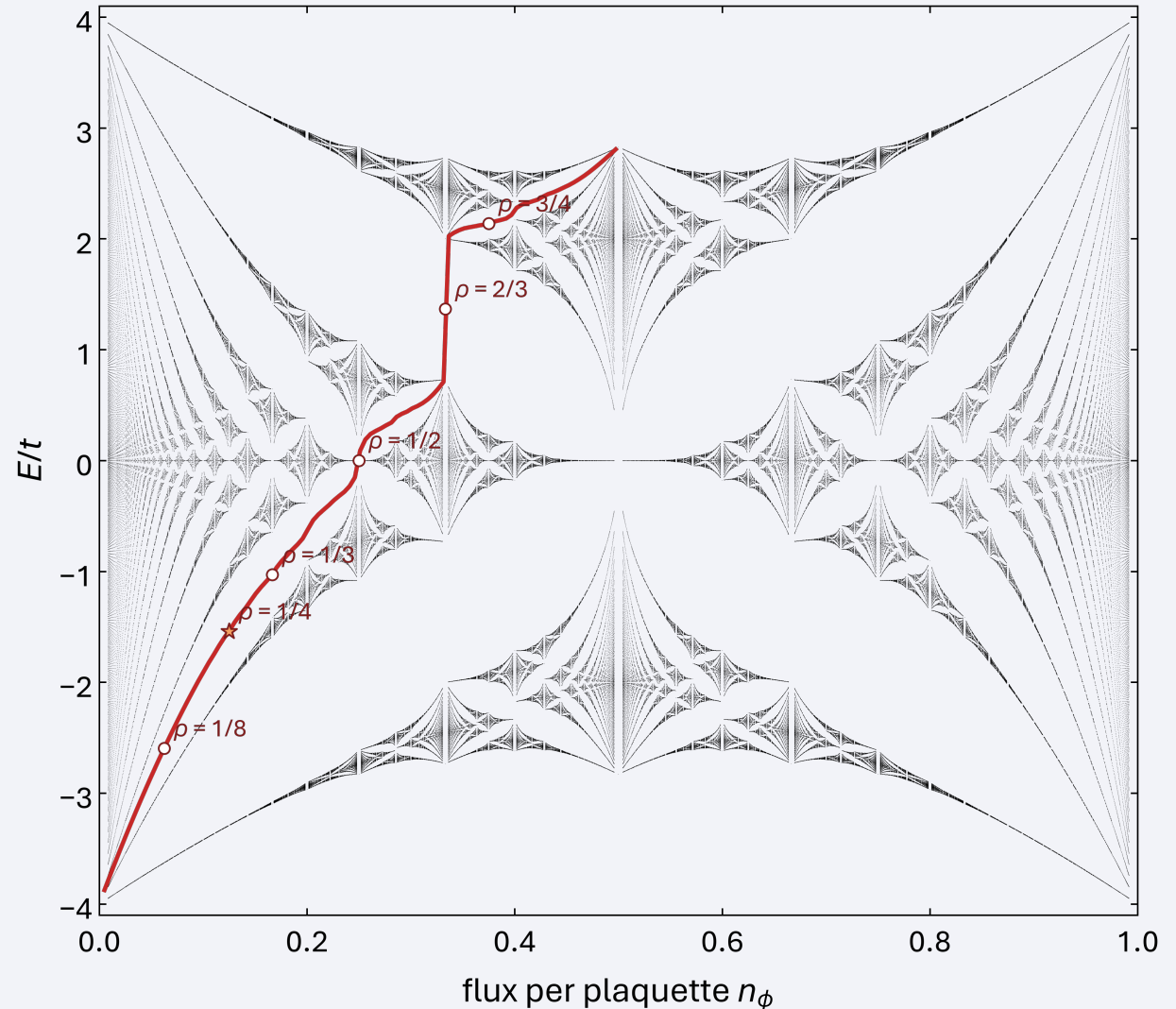


Hamiltonian symmetries, mean-field predictions

$$H = \sum_{\langle ij \rangle} T_{ij} b_i^\dagger b_j$$

$$[T_x, H] = [T_y, H] = 0$$

Mean-field predicts superfluid for $\rho < 1/2$, but numerics have only gone to 4×5 lattices



NQS Architecture

- Want to **exactly enforce** lattice symmetries
- $\psi(\mathbf{r}) = \sum_g \psi(g\mathbf{r})$
- Summation is computationally expensive, want network to output every $\psi(g\mathbf{r})$ in one evaluation
- Solution: group convolutional neural networks (GCNN)

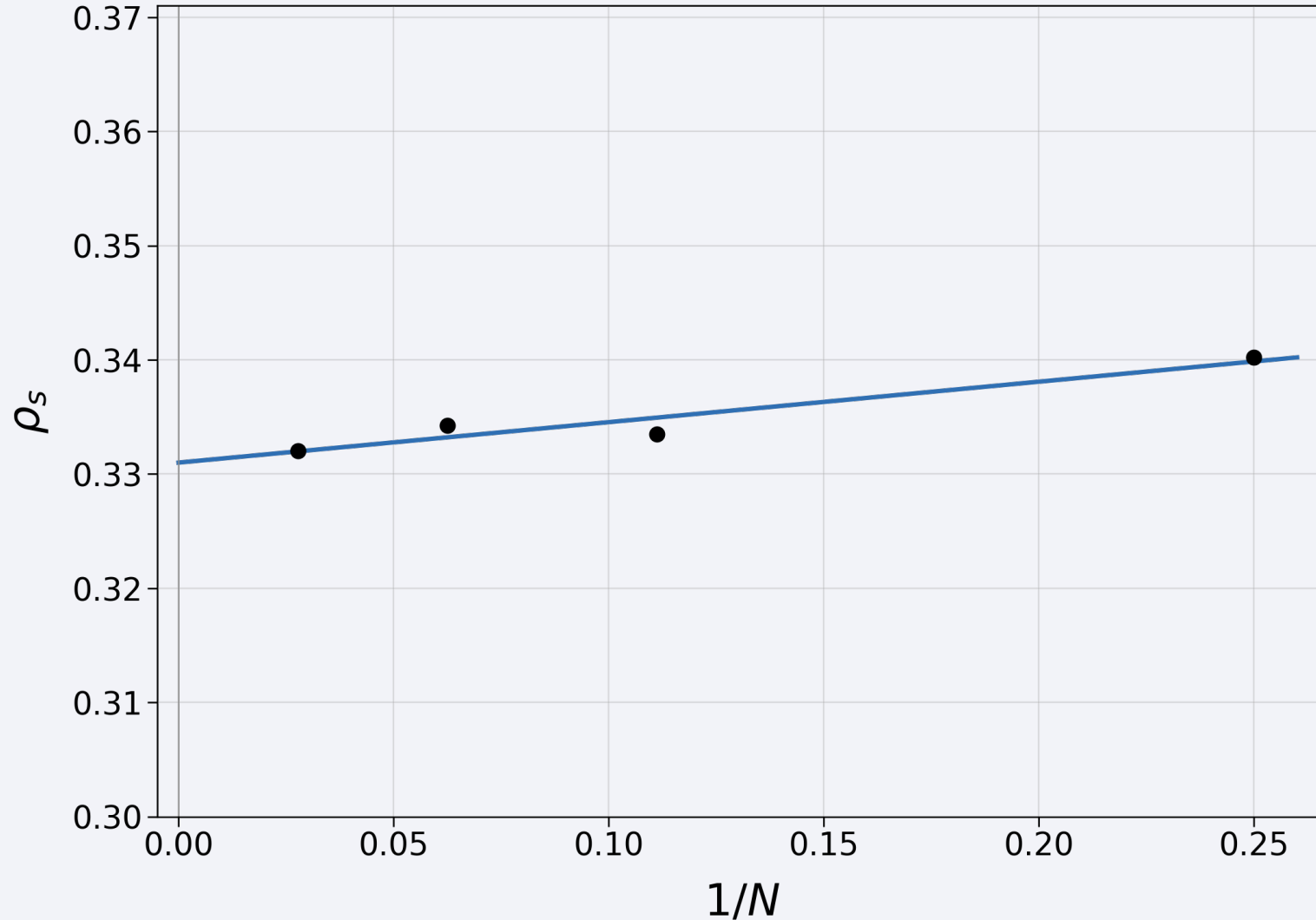
Why “long-range” interactions aren’t as severe

$$\frac{H\psi(x)}{\psi(x)} = \frac{\langle x|H|\psi\rangle}{\langle x|\psi\rangle} = \frac{\sum_y \langle x|H|y\rangle \langle y|\psi\rangle}{\langle x|\psi\rangle}$$

Normal “long-range” interactions: sum over y grows rapidly

Here: connectivity is nearest-neighbor, coefficient is
just sensitive to entire system

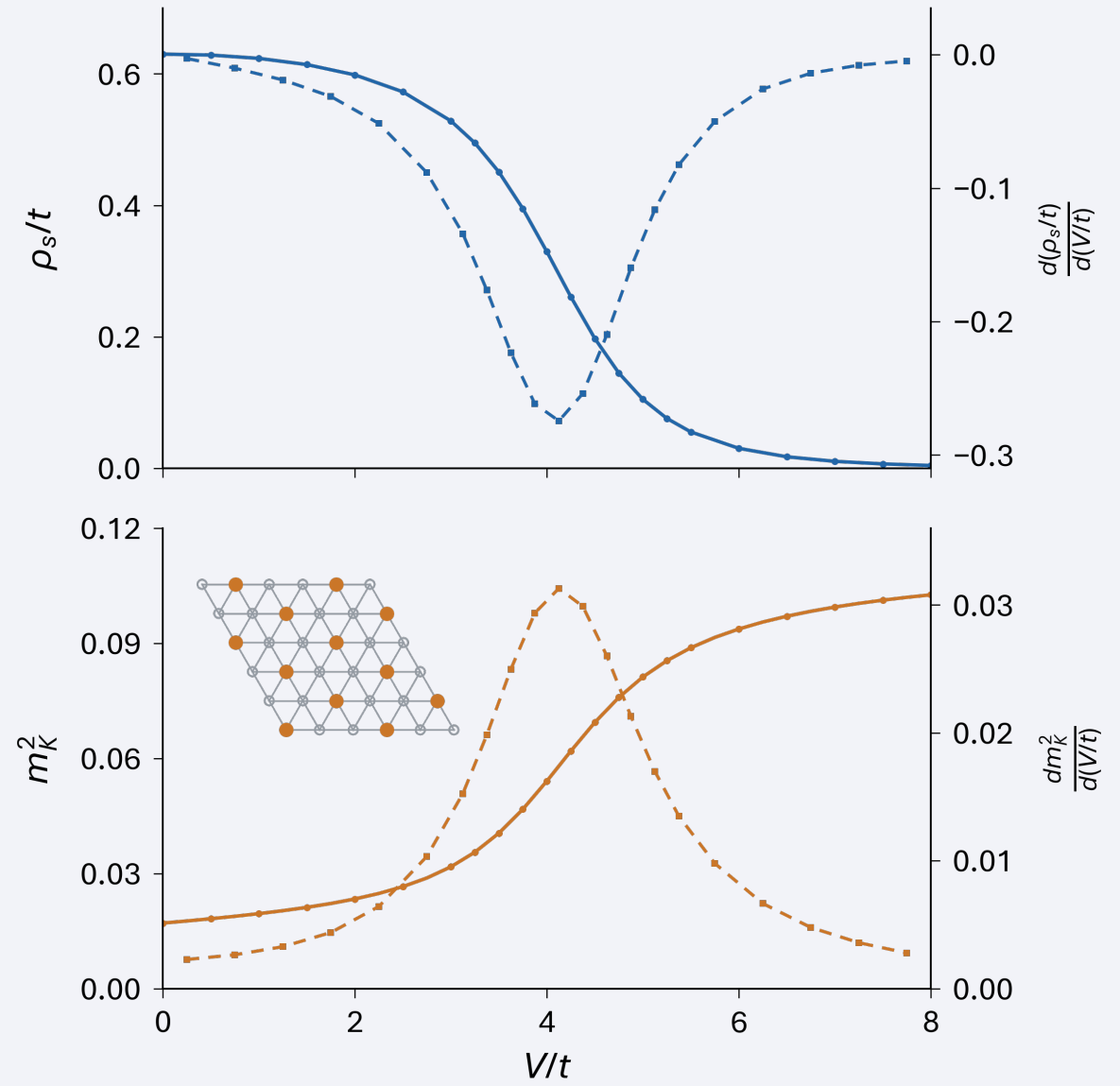
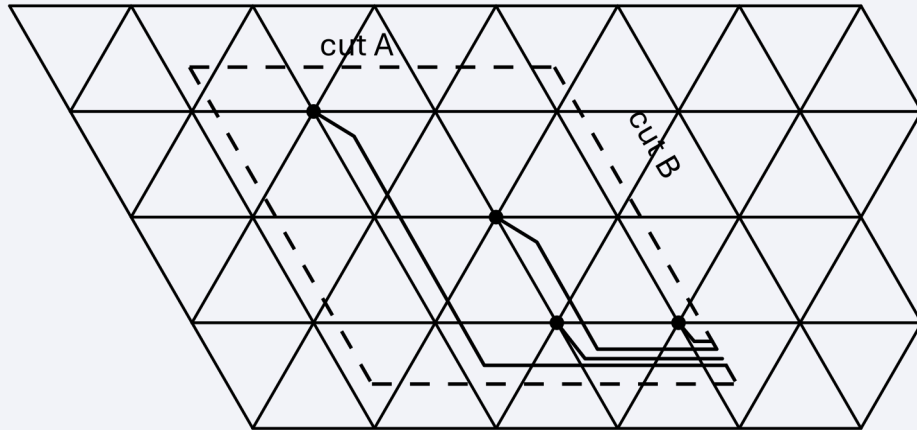
Evidence for superfluidity at $\rho = 1/4$



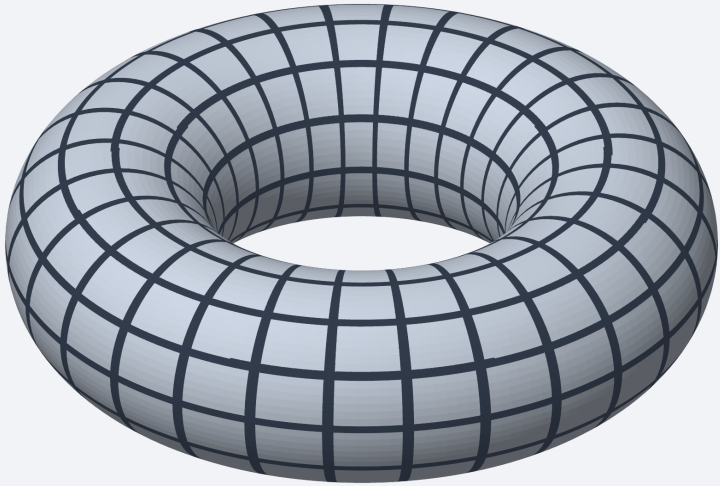
Currently $N = 4, 6, 8, 12$

Triangular lattice, semion crystal at $\rho = 1/3$

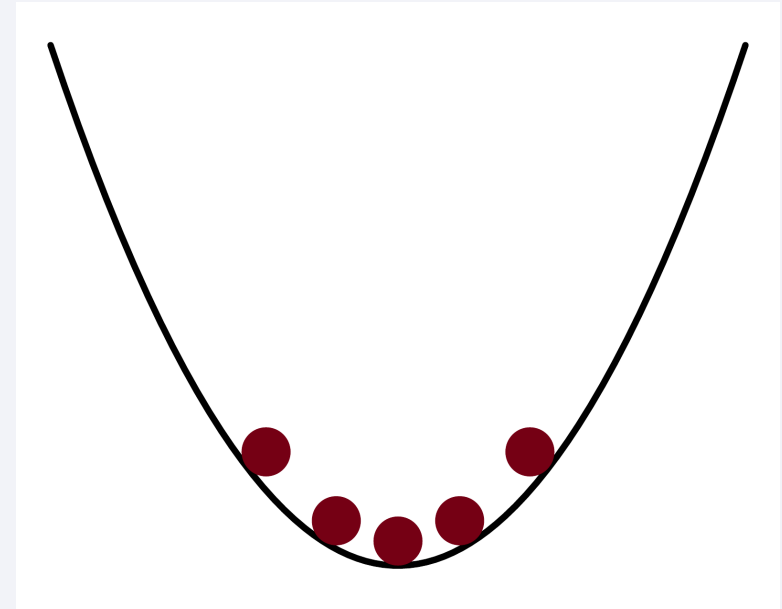
$$H = \sum_{\langle ij \rangle} [T_{ij} b_i^\dagger b_j + V n_i n_j]$$



Two systems of interest



Lattice anyons on a torus

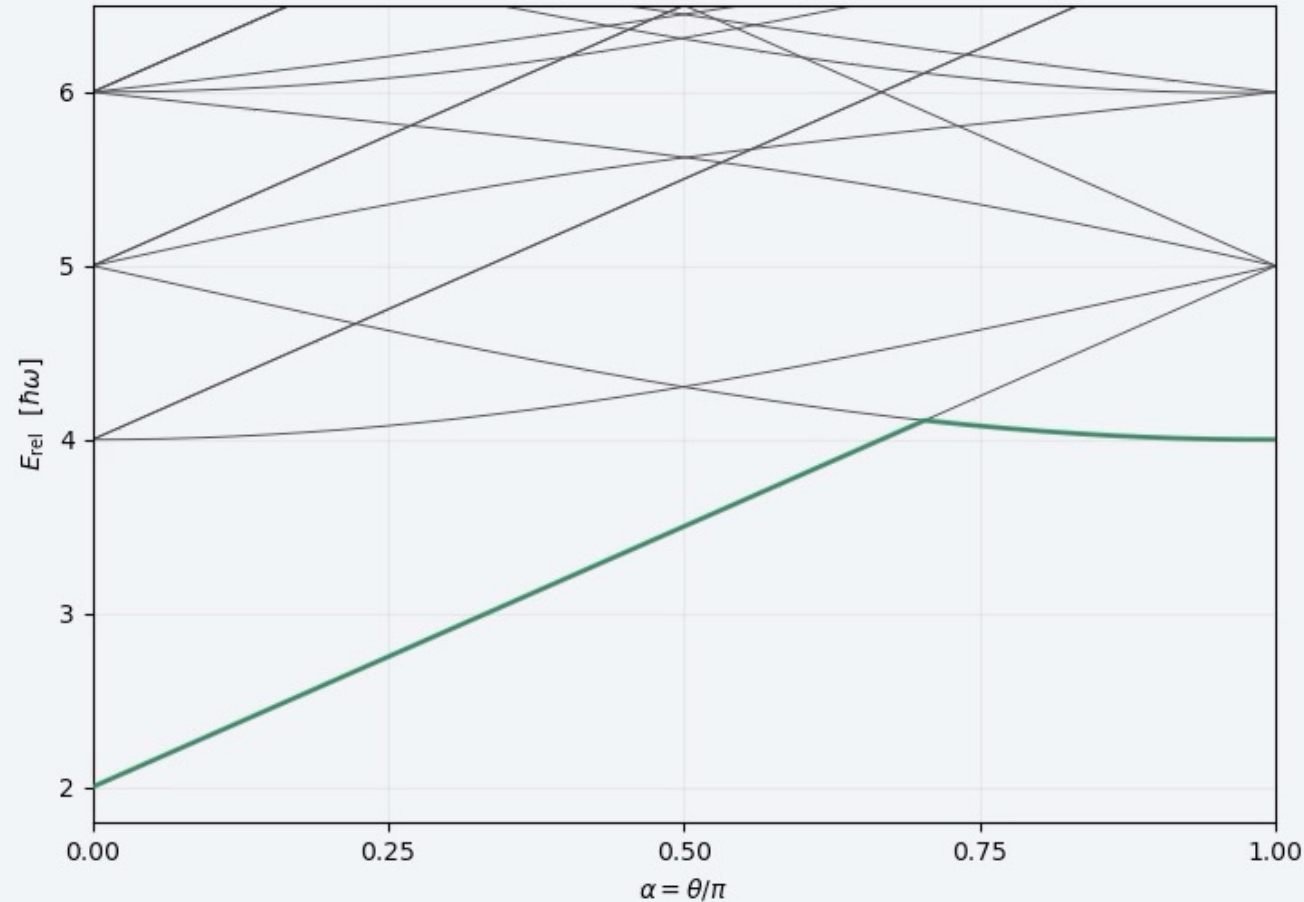


Continuum anyons in a harmonic trap

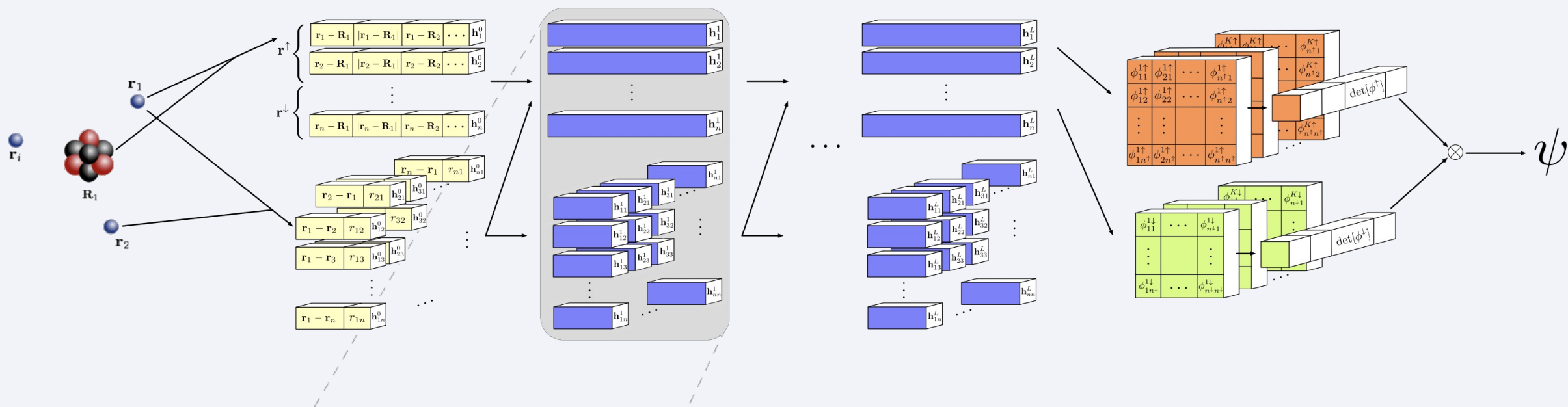
Anyons in a harmonic trap: what is known?

$$H = \frac{1}{2} \sum_{i=1}^N (-i\nabla_i + \mathbf{a}_i)^2 + \frac{\omega^2}{2} \sum_{i=1}^N |\mathbf{r}_i|^2$$

- Linear branch near the bosonic limit
- NRCFT tower of states (Nishida and Son '07)



FermiNet backbone: encoding antisymmetry with neural network orbitals



Finite local energy requires multiple short-distance channels

$$\text{As } r_{ij} \rightarrow 0, \psi(r) = f_{ij} z_{ij} r_{ij}^{\alpha-1} + g_{ij} \bar{z}_{ij} r_{ij}^{1-\alpha} + \dots \quad z \equiv x + iy$$

Required for $\frac{H\psi}{\psi}$ to be finite

Need to get non-analyticities
exactly right for training to be stable

Adopting FermiNet to correct short distances

For each orbital, $\Phi_i = w_{i0} \Phi_{i,bulk} + \sum_{j \neq i} w_{ij} f_{ij}$

Turns on when $r_{ij} \rightarrow 0$



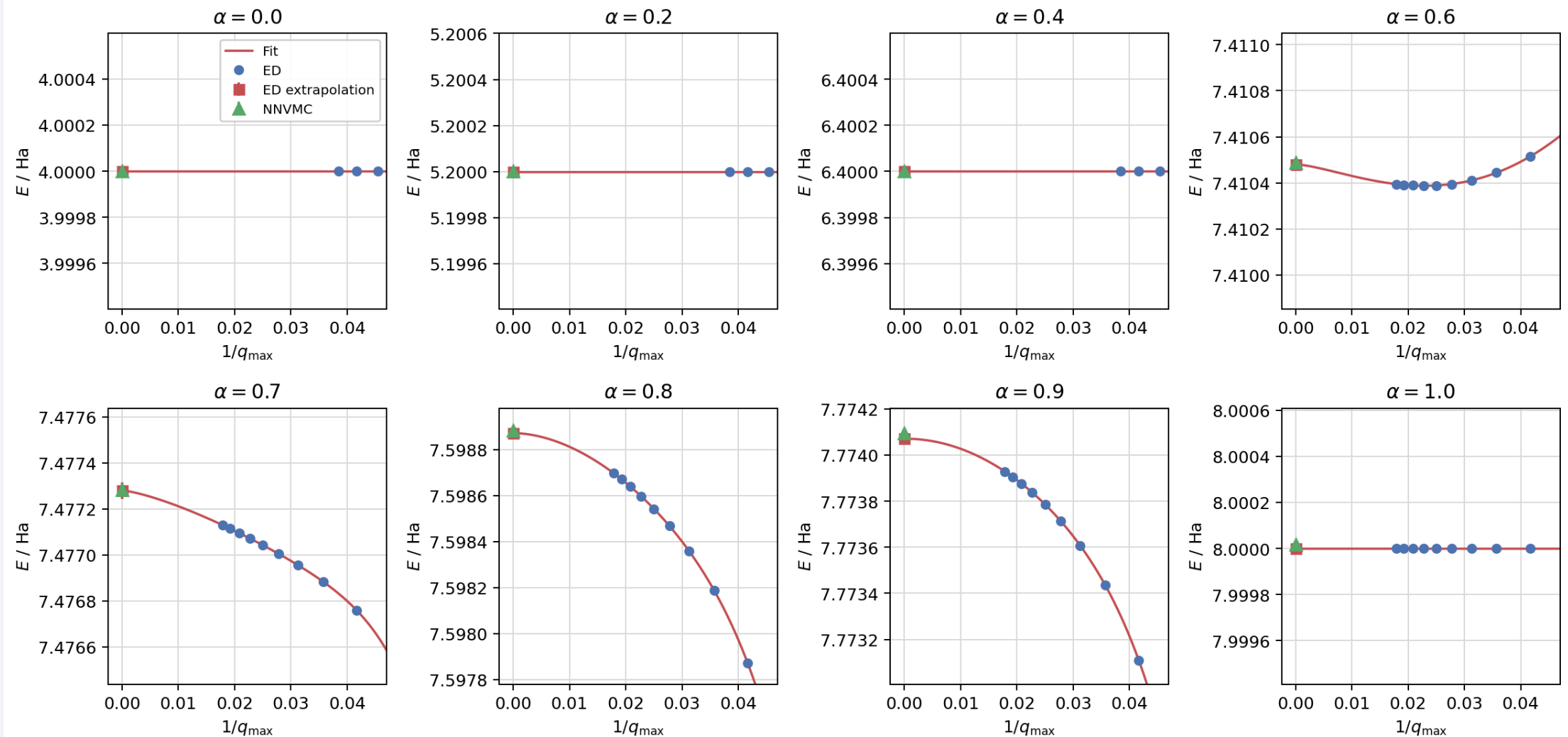
$$f_{ij} = S_{ij} + \Delta_{ij}$$

$$S_{ij} = S_{ji}, \Delta_{ij} = -\Delta_{ji}$$

$$\Delta_{ij} = V_{ij}^+ z_{ij} r_{ij}^{\alpha-1} + V_{ij}^- \bar{z}_{ij} r_{ij}^{1-\alpha}$$

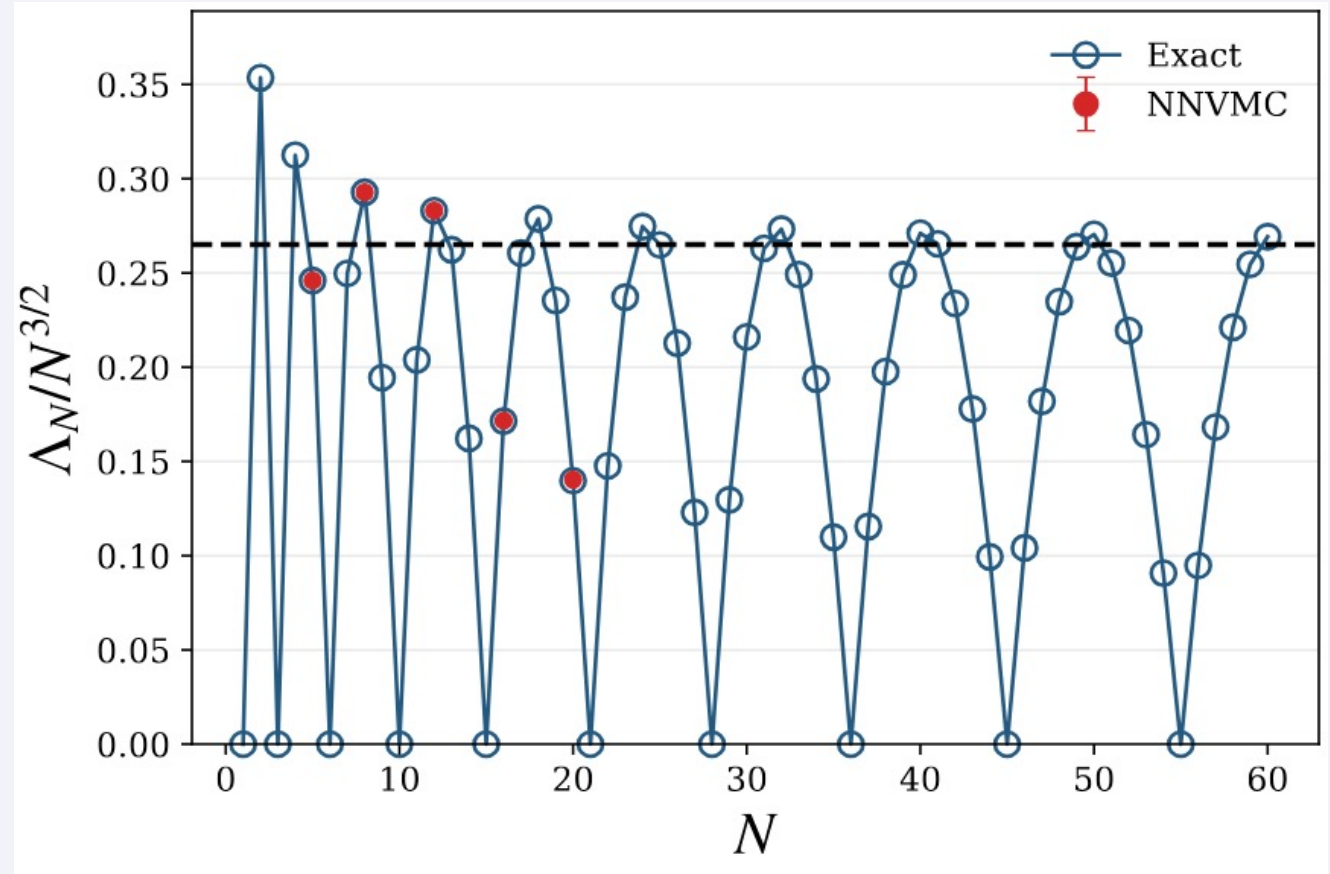
$$\begin{aligned} \det \Phi &\approx \det[S_{12} + \Delta_{12}, S_{21} + \Delta_{21}, \Phi_{3\dots N}] = \det[S_{12}, \Delta_{12}, \Phi_{3\dots N}] \\ &= z_{ij} r_{ij}^{\alpha-1} \det[S_{ij}, V_{ij}^+, \Phi_{3\dots N}] + \bar{z}_{ij} r_{ij}^{1-\alpha} \det[S_{ij}, V_{ij}^-, \Phi_{3\dots N}] \end{aligned}$$

$N = 4$ benchmarks

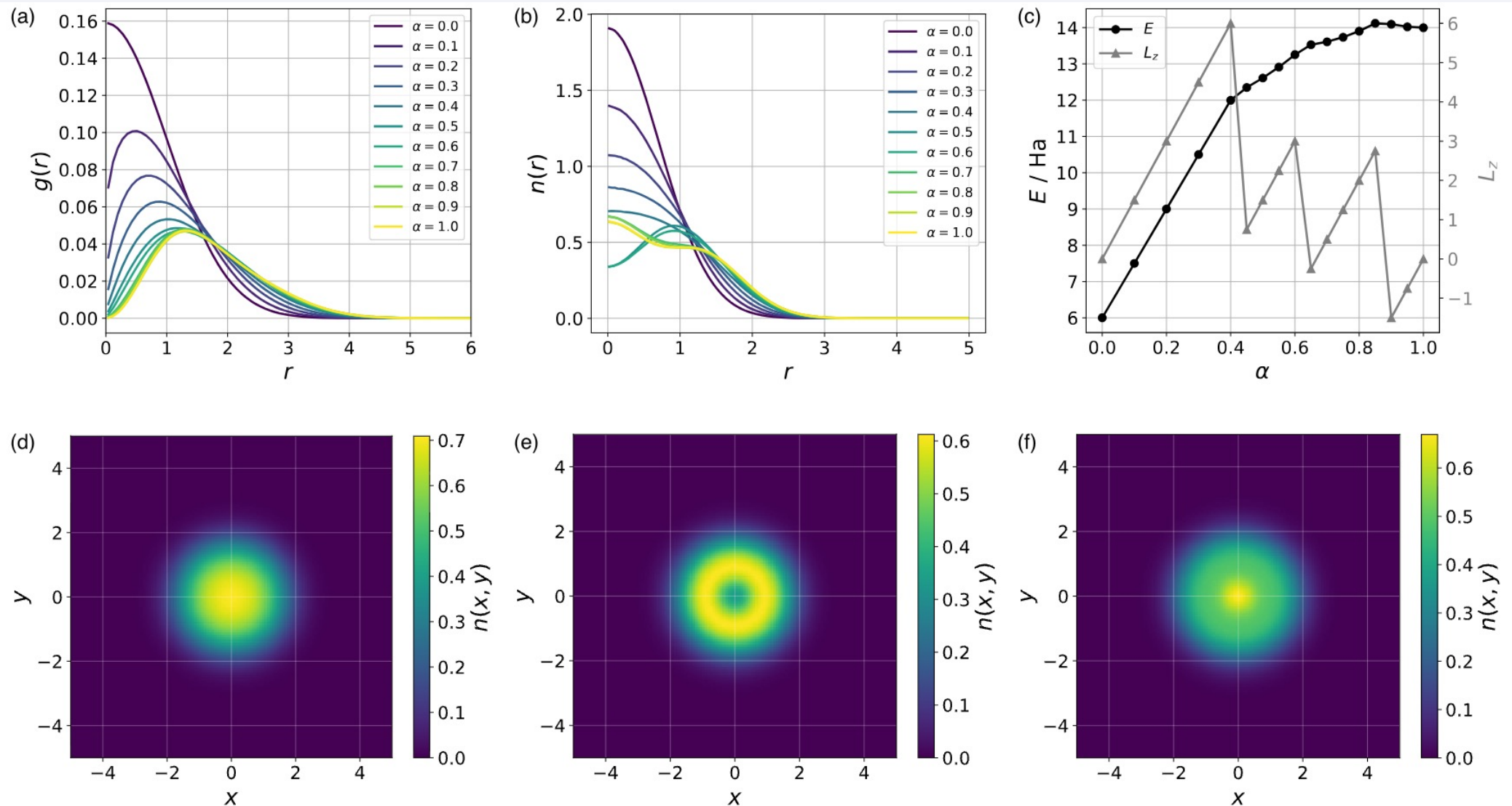


Perturbative agreement in the fermionic limit

$$E = E_F + \Lambda_N(1-\alpha) + \dots$$

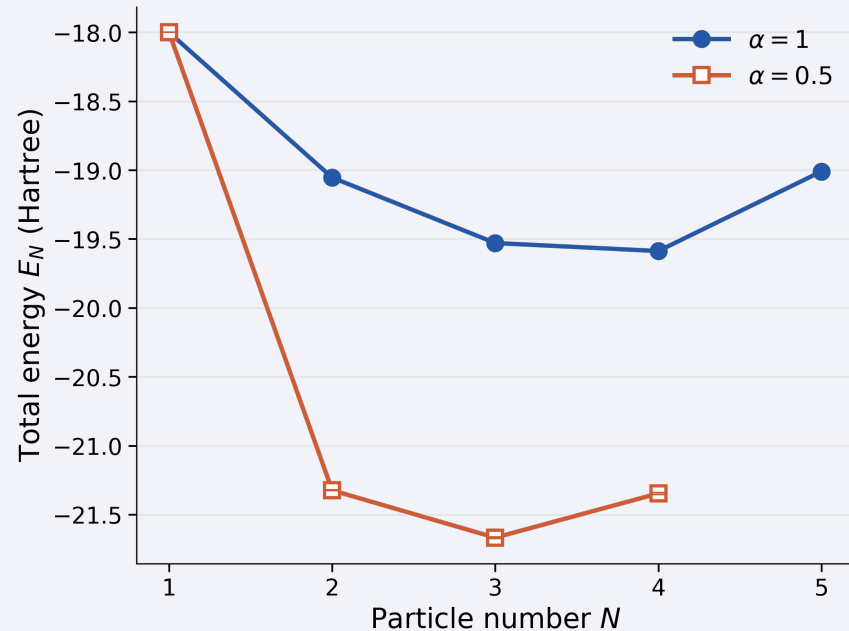


$N = 6$ results, intrinsic anyonic branch



Additional applications: anyonic quantum chemistry

$$H = -\frac{1}{2m} \sum_i [\nabla_i - i\mathbf{a}_i] - \sum_i \frac{Z}{|\mathbf{r}_i - \mathbf{R}|} + \sum_{i < j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$



Future directions

- Continuum anyons on a torus, sphere, etc
- Larger systems ($N = 20?$)
- Additional probes of superfluidity (ODLRO?)
- Additional interactions, modified dispersions, etc
- Neural wavefunctions for non-Abelian anyons